

HAT-P-37b

R-2551

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_260606 · created 2026-08-26T00:55:50+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461197.88 +/- 0.012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.017
Transit depth (Rp/Rs)^2	2.86 +/- 0.56 [%]
Orbital Inclination (inc)	89.73 +/- 1.57
Airmass coefficient 1 (a1)	2.396 +/- 0.057
Airmass coefficient 2 (a2)	-0.8 +/- 0.3
Scatter in the residuals of the lightcurve fit is	2.37 %
Best Comparison Star	#1 - [360, 452]
Optimal Aperture	2.7
Optimal Annulus	6.93
Transit Duration (day)	0.088 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.169 $\pm$ 0.017 vs 0.1394 (deviation 1.38 σ)
Duration fit vs published	0.088 d vs 0.0983 d (deviation 0.68 σ)
Mid-time O-C	23.7 min
Depth SNR	7.9
Residual scatter	2.37 %

Light curve

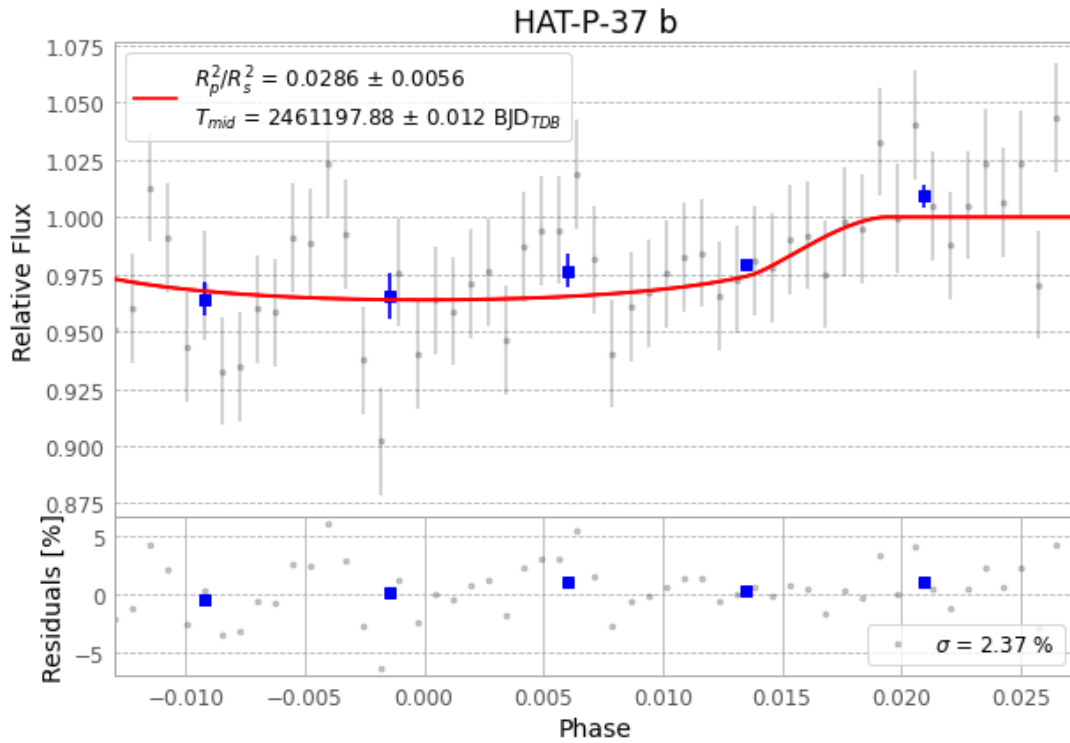


Figure 1 — FinalLightCurve_HAT-P-37 b_2026-06-06.png (SHA-256 70281d0a1304d0f1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [268, 245], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ec7e736a608a7a24...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

55 FITS frames (36 MB), HATP-37260606081217.FITS → HATP-37260606105416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-260606.log (f2cf76daf069...), FinalLightCurve_HAT-P-37 b_2026-06-06.png (70281d0a1304...), FinalLightCurve_HAT-P-37 b_2026-06-06.csv (8a3841d2a81b...)

Compute resources

Wall time 118.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 00:55Z.